# Ice Core

Season Science Lead · Vestri Dome

15 DAYS · 52 STOPS · GRADE 12 · ICECORE

## THE OPENING CARD

Vestri Dome and Skarv Camp drilled ice four hundred kilometres apart. Both cores cover one cold spell, and the two records do not agree. You are the season science lead at Vestri, so the signing is yours. In fifteen days you sign the Vestri Record: what the ice measured, which clocks turn that into dates, and where the proof stops. If the gap is really the two labs and you call it climate, every record tied to these cores carries it for years.

DAY 1 OF 15

# What the plateau has been writing down

BEFORE THE DAY — GREET

**Fourteen people, one season, and the aircraft leaves on Wednesday**

The camp is fourteen people who will be within fifty metres of each other for six weeks, and half of them arrived on the last flight. Halvorsen, the chief scientist, wants as many of them known to you as you can manage before the drill turns, because on a plateau the only thing you cannot fix later is a working relationship.

PLAN CARD 64W

First morning. Vestri and Skarv should describe the same cold spell, and their curves do not line up. Nobody can say yet whether that gap is climate or a measuring mistake. Today you fix the clock this site runs on. That means how much snow makes one year, why the hole sits on the dome, and what a year looks like in the ice.

BRIEFING 20W

Before the first drill run, the team has to know how fast this site turns snowfall into a climate record.

WHAT YOU DECIDE 18W

Establish how fast Vestri writes time, what the core can preserve, and what can already bias the comparison.

## 1.1 How thick a year is here

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · BALLPARK

**SCENE** 36W

Nadia Brandt, the field glaciologist, has the pit wall photographed and the last summer surface marked at 32 centimetres down. The snow above it weighs a third of what the same thickness of solid ice would.

**GUIDE** 68W

Five numbers, and two of them belong elsewhere: the density of liquid water, and the days of stake readings. Ask of each whether the conversion needs it. Snow and ice are the same material at different densities. So a thickness of snow is not a thickness of record. Only ice equivalent lets two sites be compared, and this site gets a tenth of what a coastal one does.

**ASK** 7W

Estimate this year's accumulation in ice equivalent.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

snow becomes ice by being compressed under the snow that falls on it · why ice keeps a record at all: burial without melting — taken as read

**VERDICT** 96W

Snow and ice contain the same water but not the same amount of air. Multiply the snow thickness by the ratio of snow density to ice density. That converts the layer to ice equivalent, the mass-based thickness two sites can be compared on. Vestri adds about 12 centimetres of ice equivalent in a year. That low accumulation buys a long record but also makes annual layers thin quickly and increases the gas–ice age difference. This rate feeds the modelled chronology. The counted years and the independent tie points are what catch the model if it drifts.

**TAKEAWAY** 15W

Accumulation is the rate at which this site writes time, expressed in conserved ice equivalent.

## 1.2 Why the hole is on the dome

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · CHOICE

**SCENE** 37W

The drill engineer, Pär Lindqvist, has the hole at 2,040 metres. On the wall is a map with 3 abandoned sites on it, all of them on slopes. Vestri Dome is the high point for 200 kilometres.

**GUIDE** 69W

Four things a dome might buy, and three of them are about the ice being cold, thick or flat. Ask of each whether it is about where the ice came from. An ice sheet flows outward from its high points. On a slope the ice at a given depth arrived from upstream, so the record mixes one place's weather with a journey. Domes pay for that in thin layers.

**ASK** 14W

What does drilling at the dome buy that a site on the slope cannot?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

ice spreads outwards from the high points of an ice sheet · accumulation rate, and what a site with little snow buys and costs — taken as read · annual layer counting, and where counting fails — taken as read

**VERDICT** 72W

Ice flows away from high points, so a core on a slope contains ice that fell upstream and travelled into the borehole. Near an ice divide the horizontal motion is much smaller. That makes the record more nearly local, and a one-dimensional chronology more defensible. It is not zero-flow ice: deep layers can still deform and move. That distinction matters later, when the simple flow model begins to fail near the bed.

**TAKEAWAY** 13W

> A dome reduces the upstream-history problem; it does not make ice flow disappear.

## 1.3 What makes one year different from the next

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · PROTOCOL

**SCENE** 33W

A metre of core from 60 metres is on the light table. The chronology lead, Ines Okonkwo, has four features marked on it and wants each one attributed before the section is cut.

**GUIDE** 63W

Four features on one metre of core, and four things that could have made them. Pair them by asking what each feature is physically made of: crystals, acid, mineral grains, or bubbles. Then ask what season each belongs to. No single marker survives the whole core, so a year is where several independent signals agree. Where only one does, it is a candidate.

**ASK** 7W

Read one annual layer off the core

**BEHIND THE BUTTON** 193W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison. The question is not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. The responses are a set to be distributed rather than a list to be sampled. So every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation. That is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

**TAKES AS READ**

snow that falls in summer sits under different conditions from snow that falls in winter

**VERDICT** 86W

No single marker survives the whole core, so a year is established where several independent signals line up. Coarse crystals form on a summer surface that stayed exposed. Conductivity spikes are acid, and volcanic acid arrives in a season. Dust is delivered by winds thousands of kilometres away and peaks with their season. The light and dark banding is bubble structure changing as each layer was compressed. Where three of the four agree, a year is a year. Where only one does, it is a candidate.

> **TAKEAWAY** 15W
>
> A year is counted from several signals that agree, not from one that looks convincing.

**END OF THE DAY** 11W

The comparison is only as good as the clock underneath it.

DAY 2 OF 15

# Every metre gets one recovery

**PLAN CARD** 40W

Second day. The drill is going back into ice that fractures on the way up, and one section has lost its depth mark. Today you decide what damage can be repaired, and what doubt has to travel with a sample.

**BRIEFING** 17W

The drill enters the brittle zone and one unlabelled section is already testing the chain of custody.

**WHAT YOU DECIDE** 15W

Recover, label, and protect the ice without adding uncertainty that can never be removed later.

## 2.1 The run through the brittle zone

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · SEQUENCE

**SCENE** 39W

The barrel is at 780 metres, inside the depth range where the ice comes up cracked. The winch, the fluid and the storage are all set for ordinary ice and Lindqvist wants the order settled before the lift starts.

**GUIDE** 64W

All five of these will happen, so ask what each one buys the next. Between about five hundred and twelve hundred metres the bubbles are at pressures the surface cannot support. Ask of each step whether it reduces the shock, lets the pressure equalise, or fixes information that is about to be lost. One of them destroys the depth record if it comes early.

**ASK** 6W

Bring a brittle section up whole

**BEHIND THE BUTTON** 169W

Why the order is graded whole. A sequence is a claim about dependency. Each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits. Partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them. That usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply. What is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility. Reversibility puts every observation that leaves the sample as it was before any that consumes it. On those boards a chronological reading finds nothing. None of the steps has to happen at a particular hour.

**TAKES AS READ**

gas trapped at depth is at the pressure of the ice around it · why ice keeps a record at all: burial without melting — taken as read

**VERDICT** 83W

Between about 500 and 1,200 metres the bubbles are at pressures the surface cannot support. The ice around them fractures as it is brought up. Each step buys the next one. A slow lift reduces the shock. Resting the core at hole temperature lets the pressure equalise gradually. Logging the breaks fixes what depth each fragment came from before anything moves. Storing it for weeks lets it finish cracking. Cutting first destroys the depth information and cuts a section that is still moving.

**TAKEAWAY** 11W

Ice recovered badly is a shorter record, not a cheaper one.

## 2.2 The metre with no label

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · CHOICE

**SCENE** 37W

A section has arrived at the line with a broken depth mark. Sigrid Holt, the core line technician, can place it to within about four metres from the run sheet. The isotope team has asked for it.

**GUIDE** 72W

Four options for a section that can only be placed within four metres. Ask of each what the depth is for. It is what turns a measurement into a statement about a year, and four metres here is thirty-odd years. Some work does not use depth at all. And note the real rule. The uncertainty has to travel with the sample, because the danger is somebody downstream who never hears about it.

**ASK** 12W

What should a section with a broken depth mark be used for?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

a measurement belongs to the depth it came from, and to no other · drilling: fluid, brittle ice and what damage costs the record — taken as read

**VERDICT** 91W

A depth is what converts a measurement into a statement about a year. Four metres here is thirty-odd years. So an isotope value from this section cannot be placed in the record. It would sit in it as a point with no time. It is still perfectly good ice for anything the depth does not enter: density, crystal fabric, how the drill is performing. The rule that matters is that the uncertainty travels with the sample. The danger is not the gap. It is somebody downstream who never hears about it.

**TAKEAWAY** 16W

> A sample with uncertain depth can still answer questions that do not require a precise age.

## 2.3 The sample that was clean until it was handled

COLD LABORATORY · A ROOM · DIAGNOSIS

**SCENE** 33W

The cold laboratory technician, Elena Cruz, has a sodium result four times higher than the run before it. She has the blank, the run order and the section's handling record on the bench.

**GUIDE** 65W

Five readings, and the quiet ones do the work. The blank before is ordinary and the blank after is high. The trim was done to specification. The replicate core holds the same year with no sodium in it. Ask of each candidate how many of the five it covers. What identifies the source is not the loud number but which of the others stayed quiet.

**ASK** 7W

Find out where a signal came from

**BEHIND THE BUTTON** 175W

Why the unremarkable readings decide it. The salient reading is what draws attention. It is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved, and which have not. A normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again. Which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation. The prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so. A partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

**TAKES AS READ**

a blank is run through the same process as a sample, with no sample in it · what a proxy is: the chain between the thing and the number — taken as read

**VERDICT** 85W

The blank before the sample is ordinary and the blank after it is high. That places the contamination inside the run rather than in the ice. The trim was done to specification, so the outside of the section is ruled out. The replicate core holds the same year with no sodium in it. So the atmosphere is ruled out as well. Every clean reading here is doing work. What identifies the source is not the loud number, but which of the quiet ones stayed quiet.

**TAKEAWAY** 15W

> A blank measures the process, so anything it shows was never evidence about the ice.

## 2.4 The level that holds the hole open

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · HOLD

**SCENE** 42W

The borehole is three kilometres deep and the ice around it is trying to close it. The fluid column is what holds it open. The level in the tower has to stay where the pressure balance needs it while the drill runs.

**GUIDE** 57W

Hold the fluid level inside the band on the tower gauge. The band narrows as the hole deepens, because the deeper the hole the less margin there is between closing it and fracturing it. The pump between the tank and the hole is your control. Every run of the winch changes the balance until you answer it.

**ASK** 8W

Hold the fluid level while the drill runs.

**BEHIND THE BUTTON** 141W

Why the hole needs holding at all. Ice at three kilometres is under enough pressure to flow, and it closes a borehole slowly and permanently. A column of fluid at close to the ice pressure holds it open. That is the only reason a deep hole can be re-entered at all.

Why too much is as bad as too little. Overpressure fractures the wall and takes fluid into the ice, which is both a lost hole and a contaminated record. The band is two-sided for that reason, and it is not symmetric in consequence.

Why the drill itself moves it. Pulling the drill out displaces less than it did going in. Chips come up with fluid on them. Cold fluid coming off the reel is denser than what is in the hole. Each of them is a rate, not a step.

**TAKES AS READ**

a column of fluid exerts pressure that grows with its depth · why ice keeps a record at all: burial without melting — taken as read

**VERDICT** 171W

Ice this deep flows, and a borehole through it closes on its own unless something holds it open. That something is the fluid column. Its pressure at the bottom has to sit close to the pressure of the ice around it. A little under and the wall creeps in. A little over and it fractures and swallows fluid. Everything in a drilling run disturbs that balance and none of it is a single step. Pulling the drill removes displacement continuously while the winch runs. Chips come up carrying fluid. Cold fluid off the reel is denser than the column it joins, and keeps sinking into it. So the pump is set to a new rate and held rather than tapped. The band narrows with depth because the two failure modes converge. The deeper the hole, the smaller the gap between the pressure that closes it and the pressure that breaks it. Both ends of that gap cost the same thing. A hole nobody can re-enter, and a season nobody can repeat.

**TAKEAWAY** 15W

A hole is held open by a balance, and a balance is held by somebody.

END OF THE DAY 14W

A clean measurement is useless if nobody can say what depth it belongs to.

# Where the count stops

**PLAN CARD** 34W

Wednesday. The layers thin with depth until the line cannot tell one year from the next. Today you mark where counting ends, and turn eighty-eight metres of firn into the mass it really holds.

**BRIEFING** 20W

There is a second trap at the top of the core: buried firn compacts, so depth is not yet time.

**WHAT YOU DECIDE** 17W

Find the end of the annually counted record and separate physical depth from age in compacting firn.

## 3.1 The depth the counting runs out

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · SWEEP

**SCENE** 54W

Snow buries snow, and each year's layer is squeezed thinner by the weight above it and the ice spreading below it. The scanner can separate two bands three millimetres apart and no closer. Okonkwo needs the depth where the annual layers thin to that limit. Past it the years cannot be counted at all.

**GUIDE** 57W

Drag the depth. The readout is the measured thickness of one year's layer there, in millimetres. Only the depths you stop at get plotted. You are not looking for a peak. You want the depth where the thickness falls to the scanner's three-millimetre limit. Sweep down the core, find that crossing, then mark the depth and commit.

**ASK** 22W

The line resolves bands three millimetres apart and no closer. At what depth does the annual layer thickness fall to that limit?

**BEHIND THE BUTTON** 155W

Why layers thin with depth. Ice under ice is compressed. An ice sheet also spreads sideways under its own weight, stretching each buried layer thinner as it goes. Near the surface a year is centimetres of ice. Two kilometres down the same year's snowfall may be a few millimetres.

What the limit costs. Above the crossing, a year is a band somebody can point at. The age of the ice is a count. Below it, two years arrive inside one scanner resolution and the count stops. The ages then come from a flow model instead. That is an inference with its own uncertainty, not a tally.

Why the crossing is a depth rather than a year. The thinning depends on the ice's own flow, so the same limit lands at different depths in different cores. It has to be measured in each one, which is why this reading is taken before any chronology is published.

**TAKES AS READ**

layers are stretched thinner as the ice below them spreads outwards · accumulation rate, and what a site with little snow buys and costs — taken as read

**VERDICT** 94W

Ice spreads outwards under its own weight, so every layer is stretched thinner as it sinks. The thinning compounds with depth. At Vestri the surface lays down about 12 centimetres a year in ice equivalent. At 2 kilometres down the same year is a few millimetres. Where the layers close to within the resolution of the measurement, the count stops. Not because the years are gone, but because two of them can no longer be told apart. Everything deeper is dated by a flow model anchored to whatever dated events can still be found.

**TAKEAWAY** 13W

A count ends where the instrument stops separating one year from the next.

## 3.2 Which year marker survives the depth

COLD LABORATORY · A ROOM · CHOICE

**SCENE** 27W

The isotope chemist, Yuki Tanabe, has 4 annual signals measured on the same stretch of core at 1,400 metres. There a year is about 26 millimetres thick.

**GUIDE** 61W

Four annual signals, and a year here is twenty-six millimetres thick. Ask of each whether the thing being measured can move through the firn. Water molecules travel through open pores for decades, and that smooths anything thinner than about twice the distance they cover. Sea salt and mineral grains do not travel. And two of these fade for other reasons entirely.

**ASK** 15W

At 1,400 metres, which signal is still able to separate 1 year from the next?

**BEHIND THE BUTTON** 197W

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**TAKES AS READ**

water molecules can move through the connected air spaces in firn · firn: snow densifying into ice, and the close-off depth — taken as read · stable isotopes as a thermometer: $\delta^{18}O$ and deuterium — taken as read

**VERDICT** 75W

Water molecules exchange and diffuse through open firn, which damps the annual water-isotope cycle before the ice is deeply buried. Particle and ion records are governed by different transport and post-depositional processes. So seasonal dust and sodium structure can still be identified after the isotope seasonality has been smoothed away. They are not perfect or immobile signals; they simply fail for different reasons. Visible banding and conductivity also have their own depth and event limits.

TAKEAWAY 15W

Different annual markers lose resolution by different processes, so chronology is strongest where several agree.

### 3.3 How much ice is packed into the firn

SURFACE & ACCUMULATION · SNOW STUDY HUT · A PERSON · BALLPARK

SCENE 43W

Brandt's density profile puts pore close-off at 88 metres. Averaged over that column, the firn is about 0.62 times as dense as solid ice. Samuel Adeyemi, the gas geochemist, needs the ice-equivalent material above close-off before he can turn it into an age.

GUIDE 52W

Do not divide 88 metres by the thickness of fresh snow. Buried firn compacts, so old annual layers are physically thinner than they were at the surface even before deep-ice flow matters. Convert the whole firn column to ice equivalent first. That keeps the conserved mass and removes the changing pore space.

ASK 10W

Estimate the ice-equivalent thickness contained in the 88-metre firn column.

BEHIND THE BUTTON 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

TAKES AS READ

the mean density ratio represents the mass in the 88-metre firn column well enough for a first estimate · accumulation rate, and what a site with little snow buys and costs — taken as read · why ice keeps a record at all: burial without melting — taken as read

VERDICT 85W

Firn gets denser as its pore space collapses. So a layer that fell as 32 centimetres of fresh snow is thinner than that by close-off. Multiply the physical depth by the mean density relative to ice. That gives the thickness the whole column would have as solid ice: about 55 metres. On Day 6 that column is divided by the annual accumulation in ice equivalent. The answer is how long the ice took to reach close-off. Dividing 88 by 0.32 would ignore centuries of compaction.

TAKEAWAY 13W

In compacting firn, physical depth is not time until density is accounted for.

### 3.4 A year, or a summer that melted

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · BELT

**SCENE** 41W

Sections come out of the trench faster than they can be logged properly. A layer that formed over a year is one thing. A layer where the surface melted and refroze is another, and both are visible on the light table.

**GUIDE** 60W

Two bins, and the question is whether the layer is a year of accumulation or a melt event inside one. An annual layer pairs a coarse summer with a fine winter and keeps its bubbles. A melt layer is clear, dense and bubble-poor, because the water filled the spaces before it refroze. Sort on the ice, not on the label.

**ASK** 12W

Send each section to the bin that says what made the layer.

**BEHIND THE BUTTON** 91W

Why the difference decides the chronology. The count of annual layers is the clock. A melt layer counted as a year makes the core older than it is at every depth below. A year missed makes it younger. Both errors propagate all the way down.

What melt does to everything else. Percolating water moves the chemistry down through the layers it passes. So a melt layer is not only a counting problem. It smears the record either side of it, which is why the site was chosen where melt is rare.

**TAKES AS READ**

snow builds up in layers and can melt and refreeze

**VERDICT** 169W

An annual layer is built by a season. Coarse crystals from summer sun. Fine wind-packed snow through winter. And air trapped as bubbles throughout. A melt layer is built by water. The surface thaws, and the water percolates down and refreezes in the pore space. What comes out on the light table is clear, dense and almost free of bubbles. Both are stripes in a core and only one of them is a year. The cost of confusing them is not local. Annual layer counting is the clock. So a melt layer counted as a year makes every date below it too old. A year missed makes every date too young, all the way to the bottom of the core. Melt also does something worse than miscount. Water moving down carries the chemistry with it. So the isotopes and the ions either side of the layer are smeared. That is why a site is chosen for how rarely its surface melts rather than for how much snow it gets.

**TAKEAWAY** 15W

A layer counted wrong moves every date below it, in the same direction, for ever.

**END OF THE DAY** 14W

A date is a method plus the depth range where that method still works.

DAY 4 OF 15

# Before the other core leaves

BEFORE THE DAY — FOLLOW

**Lindqvist walks the drill floor round once**

The drill has a rhythm and it is not written down anywhere: which valve is checked at which point in the run, which noise means the barrel is loading. Lindqvist, the drill engineer, walks it once. Stay with him and stay out of the winch path, which is the one part of this camp that will not stop for you.

PLAN CARD 42W

Thursday, and the Skarv samples fly out tonight. The two records differ by 0.9 per mil, and nobody knows if the gap is the ice or the two labs. Today you spend four runs on the one question only today can answer.

BRIEFING 17W

After tonight the two laboratories cannot put the same water on the same instrument for two years.

WHAT YOU DECIDE 19W

Use the last shared day with Skarv ice to test whether part of the disagreement belongs to the laboratories.

## 4.1 What is worth measuring before Thursday

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · VALUE

**SCENE** 46W

Eleven Skarv samples, one working day, and the isotope line can take four runs. The published records differ by about 0.9 per mil through the disputed interval. Jonas Aalto, the data manager, wants the runs that could show some of that gap belongs to the laboratories.

**GUIDE** 49W

Eleven samples, one working day, and the isotope line takes four. Open each candidate and ask what its result would change about the difference between the two records. Four runs that confirm what is already agreed have spent the day. Four that could move the disagreement have bought something.

**ASK** 16W

Which four runs should the day buy to test what the 0.9 per mil mismatch means?

**BEHIND THE BUTTON** 136W

Why the difference is the subject. Two records sit side by side, with the gap between them on a third screen. That is the whole question of the week. A run that tightens a part of the record nobody disputes cannot narrow that gap.

What a day of the isotope line is worth. It is the only instrument that can settle certain questions and it runs four samples. That makes each slot a choice against the other seven. The deadline is Thursday, not the end of the season.

Why sampling depth matters as much as sample count. Four runs clustered in one interval answer one question well. Four spread across the disagreement answer a different question. Which of them is worth more depends on whether the gap is thought to be an offset or a shape.

**TAKES AS READ**

two laboratories measuring the same quantity can differ by a fixed amount

**VERDICT** 76W

A fixed reporting offset looks exactly like a climate shift if two records are plotted without cross-calibration. The paired standards measure where the two reporting scales sit; overlapping depths test the same effect on actual ice. Here both point to the same 0.4 per mil offset. Applying it reduces the raw 0.9 per mil mismatch to about 0.5, but does not explain the remaining difference. The day therefore removes one hypothesis rather than ending the story.

**TAKEAWAY** 16W

A measurement is worth its cost when the decision changes depending on how it comes out.

## 4.2 The same water, twice

COLD LABORATORY · A PERSON · CHOICE

**SCENE** 31W

Tanabe has both groups' working standards on the bench and 30 minutes of instrument time. Skarv reports its values against its own standard, calibrated in a different laboratory four years ago.

**GUIDE** 71W

Four things a joint run might establish. Ask of each whether one instrument in one afternoon could separate it from the alternatives. An isotope value is a difference from a reference water, so it carries whatever that reference is worth. Two laboratories keep their own. A scale gap and a real site difference both appear as a constant shift. That is why one of these needs both standards on one instrument.

**ASK** 22W

The two records differ by 0.9 per mil. What does running both groups' standards on this one instrument establish about that mismatch?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

an isotope value is reported as a difference from an agreed reference water

**VERDICT** 76W

An isotope value is a ratio reported relative to a reference water. Two laboratories can maintain working standards that are each traceable to the international scale yet still carry a small offset. Running both standards on one instrument estimates that offset directly. It says nothing about either site's isotope-to-temperature slope and nothing about chronology. In this case the offset is about 0.4 per mil, leaving roughly 0.5 per mil of the original mismatch still to explain.

**TAKEAWAY** 15W

> A ratio reported against two different references is two different numbers about the same water.

## 4.3 Which differences a shared instrument removes

TRAPPED GAS · GAS LABORATORY · A ROOM · PROTOCOL

**SCENE** 31W

The gas geochemist, Samuel Adeyemi, has four ways the two records could differ written on the board. There is one working day in which both cores are in the same building.

**GUIDE** 62W

Four ways the records could differ, and four kinds of evidence. Pair them by asking what sharing an instrument can and cannot remove. A stretched timescale is a chronology problem and needs an event dated in both cores. A real climate difference is what survives after everything else is eliminated. And melting alters the outside of a section more than the inside.

**ASK** 9W

Sort what a joint run can and cannot settle

**BEHIND THE BUTTON** 193W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison. The question is not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. The responses are a set to be distributed rather than a list to be sampled. So every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation. That is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

**TAKES AS READ**

two measurements of one quantity can differ for reasons that have nothing to do with the quantity · thinning with depth: flow, and where a record runs out — taken as read · annual layer counting, and where counting fails — taken as read

**VERDICT** 90W

Only the first of these is fixed by sharing an instrument. A stretched timescale is a chronology problem and needs an event of known date present in both cores. A real climate difference is what is left once every other explanation has been removed. That is why it is established by elimination rather than measured directly. Melting in transit alters the outer ice more than the inner. So it leaves a signature in the profile across a section, rather than in the average. Four differences, four different pieces of evidence.

**TAKEAWAY** 13W

Sharing an instrument removes the differences the instrument was causing, and only those.

## 4.4 How thick a year is here — Review

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · CALLBACK · BALLPARK

**SCENE** 42W

The field glaciologist, Nadia Brandt, is back from the flank camp forty kilometres downslope with a pit wall of her own. The last summer surface there sits seventy-one centimetres down, and the snow above it is denser than anything on the dome.

**GUIDE** 70W

Five numbers, and only three of them describe the flank pit. Ask of each whether it says something about that snow as it lies. The close-off depth and the dome's own ice-equivalent figure are on the board so the two sites can be compared afterwards. They do not go into the sum. The conversion is the one you ran on the dome, on a site that is not the dome.

**ASK** 8W

Estimate the flank camp's accumulation in ice equivalent.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

snow becomes ice by being compressed under the snow that falls on it · accumulation rate, and what a site with little snow buys and costs — taken as read

**VERDICT** 93W

Choosing a site is choosing between the detail a core carries and the span it covers. That trade only becomes visible once both sites are stated in the ice their snow will become. The flank camp catches more than twice the snow and lays it down denser. So its layers stay thick enough to count far deeper than the dome's. It pays at the bottom. The same three kilometres of ice is spent in a third of the time. The oldest year in it is a few thousand rather than tens of thousands.

**TAKEAWAY** 18W

A wetter site writes finer layers and runs out sooner, and ice equivalent is what shows the trade.

**END OF THE DAY** 11W

Before comparing two records, make sure the measuring scales themselves agree.

DAY 5 OF 15

# How cold was it?

**PLAN CARD** 36W

Friday. Yesterday's shared run found 0.4 per mil of the gap inside the labs, which leaves about 0.5. Today you turn Vestri's own isotope curve into degrees, and say how much of that number is assumed.

**BRIEFING** 18W

Vestri still needs its own temperature curve, and the slope that turns a ratio into degrees is somebody's

**WHAT YOU DECIDE** 16W

Turn the isotope shift into a temperature estimate without pretending the proxy is a direct thermometer.

## 5.1 What the ratio says in degrees

COLD LABORATORY · A ROOM · BALLPARK

**SCENE** 31W

Tanabe has the isotope profile against depth. The section covering the last cold interval sits 1.8 per mil below the modern surface value. The site's calibration slope is on the board.

**GUIDE** 58W

Four numbers, and two of them belong elsewhere: a slope for a different isotope, and the years of mast overlap. Ask of each whether this conversion uses it. And note where the slope comes from. It is not a constant of physics but a fit to this site's own weather, measured over the years that have both records.

**ASK** 8W

Estimate the temperature change that isotope shift represents.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

colder air loses its heavier water molecules earlier, leaving the snow isotopically lighter · stable isotopes as a thermometer: $\delta^{18}O$ and deuterium — taken as read

**VERDICT** 86W

Water carrying the heavy isotope condenses first. Air that has cooled further has already lost more of it, so the snow falling last is the lightest. Over a useful range the ratio and the temperature move together in a straight line. The slope of that line is not a constant of physics. It belongs to this site's weather. It is measured by comparing the isotope record with the mast during the years that have both. Dividing the shift by the slope gives the change in degrees.

**TAKEAWAY** 15W

A slope converts a measurement into a claim, and it is measured rather than assumed.

## 5.2 How firmly the cooling can be stated

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · CLOUD

**SCENE** 38W

Brandt has the mast record, the calibration years and the scatter around the fitted slope. The season summary needs a statement that the interval was colder than the present. And it needs to say how sure anybody is.

**GUIDE** 63W

The band is the reconstructed cooling with its calibration uncertainty on it. The summary wants to claim the interval was more than two degrees colder. Two pieces of work are available and they act differently on the band: one moves it, one narrows it. Apply what you would fund, watching the middle and the width separately. Then say whether the claim is cleared.

**ASK** 27W

Which work would let the summary state that the interval was more than two degrees colder? Place the bars to report the mean and its one-sigma uncertainty.

**TAKES AS READ**

a fitted relationship carries a spread as well as a central value · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

**VERDICT** 87W

The reconstruction is a distribution, not a number. The slope was fitted to 11 years of overlap, and it carries the scatter of that fit. A claim that the interval was more than two degrees colder is a claim about where the whole distribution sits. It is cleared by making the distribution narrower, rather than by moving its middle. More overlap years is the only action here that adds information. The source-region correction changes the central value. It leaves the spread exactly as wide as it was.

**TAKEAWAY** 14W

Narrowing a distribution and moving it are different actions with different evidence behind them.

## 5.3 What the isotope thermometer actually measures

RECORDS & REPORTING · SCIENCE MODULE · A PERSON · CHOICE

**SCENE** 29W

Aalto is drafting the summary line and has written that the isotopes give the temperature of the last cold interval. Halvorsen wants the sentence narrowed before it goes anywhere.

**GUIDE** 66W

Four things the isotope record might be about. Ask of each where and when the ratio is actually set. It is fixed as vapour condenses, in the cloud, in the seasons that deliver most of the snow. Getting from there to a ground temperature needs the site calibration. Two of these options are separate measurements entirely, one of them taken with a thermometer in the hole.

**ASK** 8W

What does the water-isotope record most directly preserve?

**BEHIND THE BUTTON** 197W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

snow forms in a cloud and falls to a surface some distance below it

**VERDICT** 81W

The core directly stores the isotope ratio of past precipitation. That ratio changes as an air mass cools and rains out. That is what makes polar water isotopes a powerful temperature proxy. The same ratio also answers to the moisture source, the transport path, the elevation, and which seasons the snow fell in. The site calibration is the step that turns the proxy into an estimate of local temperature. The borehole thermometer measures today's ice temperature and answers a different question.

**TAKEAWAY** 20W

> Water isotopes are a temperature proxy, not a buried thermometer; the calibration and its assumptions are part of the result.

## 5.4 What makes one year different from the next — Review

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · PROTOCOL

**SCENE** 39W

The chronology lead, Ines Okonkwo, has a metre from 540 metres on the light table. It is deeper than anything counted this week. Four features on it have to be attributed before the section is released to the teams.

**GUIDE** 74W

Four features and four things that could have made them. Pair them by asking two questions of each. What the feature is physically made of. And how long whatever made it went on for. An eruption lasts a season or two. A shift in the weather lasts decades. A surface lasts a summer. Only one of these four repeats on a yearly beat, and that is the one a count can be built on.

**ASK** 14W

Attribute four features on a deeper section, and find the annual layer among them

**BEHIND THE BUTTON** 193W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison. The question is not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. The responses are a set to be distributed rather than a list to be sampled. So every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation. That is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

**TAKES AS READ**

a feature in the ice was put there by something with a duration of its own

**VERDICT** 88W

Attribution is the step before counting. It runs on what a feature is made of, and how long the thing that made it lasted. Acid arrives with an eruption and stops when the eruption does, so a pair of spikes is two events rather than two years. Sea salt is delivered by weather that shifts over decades. Coarse crystals with large bubbles are one surface that sat in the sun. Only the even banding is compression repeating on a yearly beat, and only that one can be counted.

**TAKEAWAY** 13W

A feature is a year only where whatever made it repeats every year.

END OF THE DAY 13W

A proxy becomes a temperature only through a calibration whose limits stay attached.

DAY 6 OF 15

# The air is younger than the ice

**PLAN CARD** 40W

Saturday. The gas in a bubble is younger than the ice around it, because air keeps moving through the snow until the pores seal. Today you put a first number on that gap, and find the stage that sets it.

**BRIEFING** 20W

The same depth carries two ages because the snow became old while its pores were still connected to the atmosphere.

**WHAT YOU DECIDE** 22W

Build the two clocks inside one core: the age of the ice and the younger age of the gas trapped inside it.

## 6.1 How much younger the air is

TRAPPED GAS · GAS LABORATORY · A ROOM · BALLPARK

**SCENE** 36W

The pores seal at 88 metres here. Above that the firn is open to the atmosphere, and its mean density is about 0.62 of solid ice. Adeyemi has the accumulation rate on the board beside it.

**GUIDE** 69W

Four numbers, and one of them is the same year written as a snow thickness rather than as ice. Ask of each whether the units match the others. The air keeps mixing with the atmosphere until the pores close. So the bubble is as old as the sealing, and the ice is as old as the snow. Get this wrong and a gas record is plotted centuries too early.

**ASK** 12W

Estimate the age difference between a bubble and the ice around it.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

air moves freely through connected pore spaces until they close

**VERDICT** 86W

The 88-metre firn column contains about 55 metres of ice-equivalent mass. Divide that by the annual accumulation of 0.119 metres ice equivalent. It gives roughly 460 years for the surrounding ice to reach close-off. The trapped gas is much younger because the pore space stayed connected to the atmosphere during most of that burial. Treating the gas as exactly modern at close-off is only a first-order approximation. Real firn has a lock-in zone and a spread of ages. That uncertainty is carried forward rather than hidden.

**TAKEAWAY** 21W

The gas–ice age difference is large at a dry site, and its uncertainty belongs in every comparison that uses both clocks.

## 6.2 From snowfall to sealed bubble

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CHAIN

**SCENE** 26W

Lindqvist has the borehole's own record on the wall. Five stages between a snowflake landing and air being locked away, with a measurement against each one.

**GUIDE** 55W

Put the five stages in the order a snowflake actually passes through them, from landing to being sealed in. Then name the stage that decides how much older the ice is than the air inside it. Each stage has a measurement against it, so read those rather than choosing the stage that sounds most important.

**ASK** 16W

Which stage decides how much older the ice is than the trapped gas in its bubbles?

**BEHIND THE BUTTON** 165W

Why the ice and its air have different ages. Fresh snow is open to the atmosphere, and it stays open as it compacts. Air moves in and out of that layer for decades or centuries. Only when the pores finally pinch off is the air sealed. By then the ice around it has been buried a long time.

What sets the size of the gap. The depth at which the pores close, and how long the snow takes to get there. Both depend on temperature and on how much snow falls each year. That is why the gap is hundreds of years at a cold, dry site. At a warm, snowy one it is much less.

Why it matters more than it sounds. Every comparison between a gas record and an ice record has this offset in it. Read a carbon dioxide rise against a temperature change without correcting for it. You can get the order of events wrong. That is a claim about cause.

**TAKES AS READ**

a quantity that passes through several stages is limited by one of them · gas age against ice age, and why the difference matters — taken as read

**VERDICT** 73W

Air remains connected to the atmosphere through open firn while the ice matrix ages year by year. Pore close-off is the stage that ends that exchange and therefore fixes the scale of the gas–ice age difference. Mixing is essential to the process. It does not give one sharp gas age, because close-off happens over a depth range and leaves a spread of ages. That is why the simple estimate later becomes a range.

**TAKEAWAY** 16W

In a chain of stages, one of them sets the answer and the others follow it.

## 6.3 Why the gas record has no sharp edges

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · CHOICE

**SCENE** 35W

Okonkwo has the two records from the same depths on one plot. The ice record has features a few years wide. The gas record from the same interval has nothing narrower than a few decades.

**GUIDE** 66W

Four explanations for a record with no sharp edges. Ask of each whether it is a property of the site or of the laboratory. Cutting smaller pieces would help one of them a little. The pores do not all close at once. So the air sealed in one layer is a mixture spanning the years that range covers. A wetter site closes them in a decade.

**ASK** 14W

Why is the gas record smoother than the ice record from the same depths?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

a measurement that averages over an interval cannot show anything shorter than it · stable isotopes as a thermometer: $\delta^{18}O$ and deuterium — taken as read

**VERDICT** 69W

Gas moves through connected firn and pores close over a range of depths rather than on one surface. Each extracted gas sample therefore averages atmospheric air over a span of ages before it is sealed. Higher-accumulation sites generally have smaller gas–ice age differences and narrower age distributions; dry interior sites smooth more strongly in time. Cutting a smaller laboratory sample cannot restore variation that the firn already averaged away.

**TAKEAWAY** 15W

> A record made by an averaging process cannot show a change faster than the averaging.

## 6.4 The sample that was clean until it was handled — Review

COLD LABORATORY · A ROOM · CALLBACK · CHOICE

**SCENE** 41W

The cold laboratory technician, Elena Cruz, has a calcium result twice the runs either side of it. Both blanks are ordinary and the outer centimetre was trimmed to specification. The replicate core from the second hole doubles at the same depth.

**GUIDE** 63W

Four readings of one panel, and the argument turns on what a control that did not move is worth. Ask of each option which reading it would need to have gone the other way. A blank is the process with no ice in it. So an ordinary blank speaks for everything the sample touched. A second hole speaks for everything this hole did.

**ASK** 5W

What does this panel support?

**BEHIND THE BUTTON** 180W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why a clean control is a result. A reading that stays where it always sits is easy to skip past. It is the only kind of reading that can rule a mechanism out. The loud number is consistent with several explanations at once; the quiet ones are consistent with fewer, which is what makes them decisive.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

a blank is run through the same process as a sample, with no sample in it · what a proxy is: the chain between the thing and the number — taken as read

**VERDICT** 94W

A blank is the process with no ice in it. So an ordinary blank is a positive statement about everything the sample touched on the way to the instrument. The saw, the gloves, the vials, the acid. Both of them are ordinary. The outer centimetre was trimmed, so the surface handling is out as well. What is left is a second core, recovered from a different hole, doubling at the same depth. Two independent recoveries agreeing on a number is the ice speaking, and the quiet readings are what allow it to be heard.

**TAKEAWAY** 14W

A control that did not move is a measurement, not the absence of one.

END OF THE DAY 11W

One depth in an ice core can carry two different clocks.

# What the wind brought

**PLAN CARD** 59W

Sunday. The cold spell is dusty, salty and cut by one sharp sulphate spike. All of it blew in from somewhere else, and none of it means only one thing. Today you separate how much of a signal there is from what caused it. Then you decide which chemistry is worth buying if the real problem is the clock.

**BRIEFING** 18W

Dust, sea salt, and a volcanic sulphate spike offer clues that do not come from the isotope thermometer.

**WHAT YOU DECIDE** 16W

Use chemistry as evidence about transport, source, and chronology without turning concentration into a single cause.

## 7.1 The number that is a difference

COLD LABORATORY · A ROOM · BALLPARK

**SCENE** 35W

Cruz has the raw ratio of heavy to light oxygen for a sample and the same ratio for the standard water. The record is plotted in parts per thousand and the raw numbers are not.

**GUIDE** 62W

Four numbers, and one of them is the site temperature slope, which belongs to a later step. Ask of each what part of the convention it supplies. Ratios differ in the fourth decimal place, so nobody reports them as ratios. The scale is a comparison with one agreed water, set so that the standard itself reads zero. Polar snow is always negative.

**ASK** 11W

Estimate the isotope value of this sample in parts per thousand.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

an isotope ratio is the amount of a heavy isotope divided by the amount of a light one

**VERDICT** 100W

Isotope ratios differ from one another in the fourth decimal place, so nobody reports them as ratios. The convention is to divide the sample's ratio by the ratio of one agreed water. Then subtract 1 so that the standard itself reads 0. Then multiply by 1000 so the numbers are convenient. What comes out is a difference in parts per thousand. It is always negative for polar snow. Every stage of the journey from the ocean has already removed some of the heavy isotope. The scale is a comparison, and it only means anything if everyone uses the same water.

**TAKEAWAY** 20W

An isotope value is a comparison with an agreed water, which is why every one of them is negative here.

## 7.2 Four times the dust

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · CHOICE

**SCENE** 37W

Through the cold interval the dust concentration is four times the modern value and the grains are the same size distribution as today's. Okonkwo has the sodium record beside it, up by half over the same depths.

**GUIDE** 63W

Four readings of four times the dust, and three of them name a cause. Ask what a concentration is made of. It is an amount of dust divided by an amount of snow. Three separate things move it: how much was released, how well it was delivered, and how much snow fell to dilute it. Accumulation drops in cold intervals on its own.

**ASK** 11W

What is the most defensible reading of four times the dust?

**BEHIND THE BUTTON** 197W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

fine dust reaching this plateau has travelled thousands of kilometres

**VERDICT** 89W

A concentration is an amount of dust divided by an amount of snow. So it responds to three separate things. How much dust the source regions released. How efficiently the winds delivered it. And how much snow fell here to dilute it. Accumulation at this site drops in cold intervals, which raises concentration on its own. Separating the three needs the flux — concentration multiplied by accumulation — and the grain size and chemistry to identify the source. The concentration alone establishes that the delivery changed, and no more.

**TAKEAWAY** 21W

> A concentration in the ice is a product of what was produced, what was carried, and how much snow diluted it.

## 7.3 What the chemistry budget should buy

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · SCIENCETANK

**SCENE** 24W

Halvorsen has one chemistry line, six weeks, and four proposals from three groups. The depth–age scale below 1,600 metres is the season's stated objective.

**GUIDE** 85W

The section below 1,600 metres carries no dated horizon and the layer count stopped above it. Two eruptions of known date are expected somewhere in that depth range. Sulphate and conductivity find candidate horizons; tephra shard chemistry is what puts a name and a date on one. The modern pollution record is a genuine result and bears on nothing in the timescale. The sodium section has been measured twice and both runs agree inside the analytical error. There is one chemistry line and six weeks.

**ASK** 7W

Fund the chemistry that dates the core

**BEHIND THE BUTTON** 188W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all. The last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong. Funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution. They are a decision not to decide, taken with somebody else's money and deadline. The floor is what forces the panel to say something.

**TAKES AS READ**

an eruption of known date leaves a signature in ice all over the world · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

**VERDICT** 74W

The season's stated objective is a depth–age scale below the counted section. Only the volcanic work bears on it. Sulphate and conductivity find the candidates. The tephra chemistry turns a candidate into a named eruption with a date on it. Neither is worth much without the other. The pollution record is a real result for a different question. And a third run on ice that already has two agreeing measurements buys nothing at all.

**TAKEAWAY** 22W

> Work that could change what the season can claim is worth more than work that adds detail to what it already claims.

### 7.4 The run through the brittle zone — Review

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · SEQUENCE

**SCENE** 39W

The core line technician, Sigrid Holt, has a section from 1,150 metres on the light table. Four teams are down on the request sheet for a piece of it. Nothing on the sheet says which of them goes first.

**GUIDE** 63W

Five operations, all of which will happen to this section. Nothing here has to happen at a particular hour, so reading the rail as a timetable finds nothing. Ask of each how much of the section it spends. Whether it could be done again next season. Whether it leaves the inner core whole. Or whether the ice is gone once it is finished.

**ASK** 11W

Order the work on a section that cannot be drilled again

**BEHIND THE BUTTON** 169W

Why the order is graded whole. A sequence is a claim about dependency. Each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits. Partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them. That usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply. What is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility. Reversibility puts every observation that leaves the sample as it was before any that consumes it. On those boards a chronological reading finds nothing. None of the steps has to happen at a particular hour.

**TAKES AS READ**

a length of ice removed from a core cannot be put back · why ice keeps a record at all: burial without melting — taken as read

**VERDICT** 99W

Ice gets one reading. So the order of work on a section is decided by what each operation costs, rather than by when anybody wants it. A photograph can be taken again next season. A non-contact scan reads the outside and leaves everything. A trim gives up the skin that handling and drill fluid have already reached, and keeps the core. A melt turns a length of ice into a number and nothing else, and a crush ends the section. Run that order backwards and no time is lost — the measurements the earlier steps would have made are.

**TAKEAWAY** 17W

Where a sample cannot be replaced, the order of work is set by what each step spends.

END OF THE DAY 15W

A chemical signal is strongest when you know which link in the chain it constrains.

DAY 8 OF 15

# The year written in both cores

BEFORE THE DAY — HUNT

**Six core boxes on the manifest, four in the trench**

The aircraft takes what is boxed and labelled, and two boxes from last week are not where the manifest says. Until they are found nobody can say what is flying out and what is staying another year, and ice that stays another year is ice with another year's temperature history on it.

**PLAN CARD** 34W

Monday. A sulphate spike at 1,642 metres has volcanic glass in it. Today you name the eruption, say which of the two clocks it fixes, and spend the one clear day before the storm.

**BRIEFING** 15W

Glass shards at 1,642 metres may give both cores a date neither flow model produced.

**WHAT YOU DECIDE** 15W

Use a volcanic horizon to anchor the ice chronology, while keeping the gas chronology separate.

## 8.1 The spike at 1,642 metres

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · DIAGNOSIS

**SCENE** 35W

One sulphate spike, two candidate eruptions of similar age, and a set of glass shards recovered from the same centimetre. Okonkwo has the shard chemistry, the layer count and the modelled age on the bench.

**GUIDE** 59W

Five readings, and only one of them is independent of the timescale. Ask of each whether it could tell two eruptions ninety years apart from each other. The modelled age cannot: its own uncertainty is wider than the gap. Sea-salt sulphate does not arrive with volcanic glass. And the layer count stopped four hundred and sixty metres higher up.

**ASK** 3W

Identify the horizon

**BEHIND THE BUTTON** 175W

Why the unremarkable readings decide it. The salient reading is what draws attention. It is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved, and which have not. A normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again. Which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation. The prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so. A partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

**TAKES AS READ**

volcanic glass carries the chemical composition of the volcano that erupted it

**VERDICT** 87W

The shard chemistry is the only reading here that is independent of the timescale. It is the one that discriminates. Glass composition is a fingerprint of a particular magma and a particular volcano. The modelled age cannot choose between 2 candidates 90 years apart because its own uncertainty is larger than that gap. Sea-salt sulphate does not arrive with volcanic glass. And waiting for the layer count is waiting for something that stopped 460 metres higher. That is the reason a dated horizon matters here at all.

**TAKEAWAY** 21W

> A horizon is identified by what is in it, not by the date it would be convenient for it to have.

## 8.2 What a dated ash layer fixes

TRAPPED GAS · GAS LABORATORY · A PERSON · CHOICE

**SCENE** 27W

Adeyemi has the eruption date and the depth it sits at. Beside them are the carbon dioxide measurements from the bubbles in the same section of core.

**GUIDE** 69W

Four things a dated ash layer might fix. Ask of each which of the two clocks in the core it belongs to. Ash falls on the surface and is buried with the snow. The air beside it was sealed centuries later and lower down. So a tie point fixes one clock and leaves the other to a calculation. That is why the age difference has to be estimated first.

**ASK** 6W

What does the dated horizon fix?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

volcanic ash settles out of the atmosphere within a year or two of an eruption

**VERDICT** 69W

Volcanic material is deposited with the snow and therefore dates the ice matrix at that horizon. The gas in bubbles from the same physical depth was sealed later and is younger than the surrounding ice. The volcanic tie point is thus an independent constraint on the ice clock, not a direct gas-age tie point. A separate atmospheric marker—such as a methane feature shared between cores—can test the gas clock.

**TAKEAWAY** 19W

A tie point dates the material it is in, and anything else has to be reached by a calculation.

## 8.3 One clear day, three jobs

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · TRIAGE

**SCENE** 30W

Brandt has a forecast that gives her one clear day and then four of blowing snow. Three jobs are outstanding and each takes most of a day on its own.

**GUIDE** 59W

Four jobs, all of them a day's work. Ask of each whether it could be done in four days of blowing snow, and whether the storm changes what it measures. The pit is beside the hut and the mast is on a fixed line. One of these runs two kilometres out and measures a surface that drifting snow rearranges.

**ASK** 6W

Which job takes the clear day?

**BEHIND THE BUTTON** 193W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance. It is the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them. Which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it. A step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe, all of them eventually happen. What is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on. That is worth reading even when the choice was right. The ordering behind the first place is the rest of the answer.

**TAKES AS READ**

a measurement that cannot be repeated should be taken when it can be taken · why ice keeps a record at all: burial without melting — taken as read

**VERDICT** 88W

Two of these can be done in bad weather and one cannot. The pit is beside the hut, and the mast is a short walk on a fixed line. Both survive four days of blowing snow with nothing worse than discomfort. The stake array runs two kilometres out and needs visibility to level against. More to the point, blowing snow redistributes the surface the stakes are measuring. A reading taken after the storm is measuring something different. The shallow core can be re-drilled in any weather at all.

**TAKEAWAY** 13W

Order the work by what becomes impossible, not by what is most important.

**END OF THE DAY** 16W

An independent tie point is valuable because it can test the clock rather than inherit it.

DAY 9 OF 15

# Below the last count

**PLAN CARD** 36W

Tuesday. The drill is past 2,400 metres, well below the last year anybody could count. Today you check the thinning model against what the line really sees, and pick the one uncertainty worth the time left.

**BRIEFING** 18W

Below the last counted layer every age comes from a flow model, and its assumptions start to show.

**WHAT YOU DECIDE** 14W

Show why the deepest chronology is model-dependent and decide which uncertainty is worth reducing.

## 9.1 A year at 2400 metres

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · BALLPARK

**SCENE** 33W

The ice is 3,010 metres thick here and the barrel is at 2,400. At the surface a year is about 0.119 metres of ice equivalent. Lindqvist wants to know what one is now.

**GUIDE** 65W

Four numbers, and one of them is the close-off depth, which belongs to the firn rather than to the flow. Ask of each whether it describes the surface, this depth, or the whole sheet. The thinning depends on how much ice is left below, so what matters is the distance to the bed. That is what packs a hundred thousand years into the bottom kilometre.

**ASK** 14W

What does the simple thinning model predict for one annual layer at 2,400 metres?

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

ice spreads outwards under its own weight, stretching the layers below · correlation between two records is not attribution — taken as read

**VERDICT** 95W

The simple Nye-style picture makes annual-layer thickness fall in proportion to height above the bed. There are 3,010 metres of ice here and the sample is at 2,400. The model predicts about 24 millimetres for a year that began as 119 millimetres at the surface. The measured deep layering is much thinner. That mismatch is not a reason to throw away the core. It is evidence that uniform vertical strain is a poor model near the bed. Deep ages therefore need a more realistic flow model constrained by independent horizons, with model uncertainty stated explicitly.

**TAKEAWAY** 17W

A simple model is useful when its failure tells you which assumption the deep chronology cannot hide.

## 9.2 Which width the answer is made of

RECORDS & REPORTING · SCIENCE MODULE · A PERSON · PROPAGATE

**SCENE** 31W

Aalto has the dust flux from the cold interval and four quantities behind it, each with its own uncertainty. There is time to improve one of them before the season ends.

**GUIDE** 56W

The panel shows an error budget: each input's own width, times the power it carries in the calculation, contributes a bar. Read the bars rather than the inputs. There is time to improve one quantity before the season ends. So buy the measurement that shortens the tallest bar. Not the one that is easiest to improve.

**ASK** 12W

Which measurement should the remaining time buy to narrow the dust flux?

**BEHIND THE BUTTON** 127W

How the widths combine. In a product of quantities, the fractional uncertainties add in quadrature, each weighted by its exponent. A quantity raised to a power contributes that much more. One measured badly can dominate even with an exponent of one.

Why the largest quantity is not the widest one. A big number measured precisely contributes little; a small one measured roughly can carry the whole budget. The bars are what separate those two. The intuition to be resisted is the one that goes by the size of the number.

Why improving the wrong term is worse than doing nothing. It spends the last of the season and moves the total uncertainty by almost nothing. And it produces a report that looks more careful than it is.

**TAKES AS READ**

a quantity multiplied into a result carries its own uncertainty into that result

**VERDICT** 83W

A term enters the result raised to some power. Its contribution to the width of the answer is that power multiplied by its own fractional uncertainty. The grain diameter is cubed, which makes it look like the term to fix. But it is known to 5%, so it contributes 15. The accumulation rate enters once and is known to 30, so it contributes twice as much on its own. Ranking by exponent is a reasonable habit and it picks the wrong measurement here.

**TAKEAWAY** 24W

A term's contribution is its own width multiplied by the power it is raised to. The largest power is not always the largest contribution.

## 9.3 The last hundred metres

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CHOICE

**SCENE** 32W

Holt has core from the bottom of the drilled section. The layers are folded back on themselves in places and the isotope profile has reversals in it that no year could produce.

**GUIDE** 61W

Four options for folded basal ice. Ask of each what has been lost and what has not. The material is unchanged; the order is gone. So anything where depth means time is ruled out, and anything an unordered sample can answer is not. Note also what has to travel with the data, because the next person will not see the fold.

**ASK** 9W

What is the folded basal section still good for?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

ice near the bed has flowed over rough ground for a long time

**VERDICT** 98W

Ice near the bed has been dragged over rough ground for a long time, so layers are folded, sheared and sometimes overturned. The material is unchanged and the order is gone. That rules out anything that depends on depth meaning time. A temperature history, a layer count, a rate of change. It does not rule out what an unordered sample can still answer. What gases are present. How much air the ice holds. How old the oldest ice down there is. The disturbance has to travel with the data, because the next person will not see the fold.

**TAKEAWAY** 12W

Ice can be perfectly good material and still hold no readable order.

## 9.4 What the saw is cutting for

COLD LABORATORY · A ROOM · SPOT

**SCENE** 43W

Sections come to the saw with a cutting list, and the list is set by whichever analysis has priority. The chief scientist changes that priority as the season goes. After the gas request. After the dating problem. After the aircraft date is confirmed.

**GUIDE** 51W

Cut the sections the current priority wants and leave the rest in the rack. The priority on the cold-room board is changed during the day and nobody announces it. What is scored is the sections either side of a change, because a section cut for the wrong analysis cannot be uncut.

**ASK** 20W

Cut to the priority on the board, keep watching it, and keep the core log matching what leaves the saw.

**BEHIND THE BUTTON** 100W

Why the priority changes. Early the season is about the chronology, so the dating sections come first. When the gas laboratory's slot is confirmed the priority becomes anything for gas. Gas needs the core's centre and cannot use what has been near the wall. Before the aircraft it becomes anything that has to fly.

Why this is irreversible in a way most work is not. Every other mistake here can be repeated. A cut cannot. The section is gone. The depth it came from cannot be re-drilled. What was cut for one analysis is unavailable to the others for ever.

**TAKES AS READ**

a core section can only be cut up once

**VERDICT** 134W

Three priorities run across the day and each wants a different part of the rack. Anything for the dating work, then anything for gas, then anything leaving on the aircraft. A section can answer two of those at once, which is what makes the change cost something real. What the panel scores is the window either side of each change. Most of the rack is wanted by neither priority, and is correctly left alone by somebody who has read nothing. And this is the format's sharpest case, because a cut cannot be undone. Everything else on the plateau can be repeated with more time or more fuel. A section cut for isotopes is not available to the gas laboratory ever again. The ice at that depth cannot be drilled a second time this season.

**TAKEAWAY** 14W

The cost of a withdrawn instruction is highest where the work cannot be repeated.

**END OF THE DAY** 15W

The deepest result is often the one with the longest chain of assumptions behind it.

# The disagreement that survived the lab test

**PLAN CARD** 37W

Wednesday. The easy answer is gone: both labs are on one scale, and about 0.5 per mil still stands. Today you map what the two records quietly share, and put the rest of the gap into degrees.

**BRIEFING** 18W

What is left could be climate, or it could be two age scales resting on the same assumption.

**WHAT YOU DECIDE** 18W

Put Vestri and Skarv on one measurement scale and identify which parts of their chronology are genuinely independent.

## 10.1 What both records rest on

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · TRACE

**SCENE** 30W

Five pieces of the comparison are on Aalto's screen, each one computed from something. Okonkwo wants to know which of them would still stand if the flow model were wrong.

**GUIDE** 53W

Open each piece of the comparison to see what it was computed from. Keep the ones that would still stand if the flow model were wrong, and untick the ones that run through it. Then name the shared dependency. A piece that survives is worth more than the number attached to it suggests.

**ASK** 12W

Which parts of this comparison would survive the flow model being wrong?

**BEHIND THE BUTTON** 138W

What a flow model does here. It converts depth into age by describing how the ice has thinned and moved since it fell. Almost every quantity that carries a date has passed through it. That is what makes it the most consequential assumption in the building.

Why some things survive it. Anything measured against a fixed marker is a claim about ice rather than about the model. An ash layer, a known event, a directly counted section. Those hold whatever the model does, and they are how the model itself gets tested.

Why the question is asked this way round. Nobody expects the flow model to be wrong outright. The question is which conclusions are hostages to it. The flow model is revised every few years. When it is, the group knows exactly what has to be redone.

**TAKES AS READ**

two results computed from one assumption fail together if that assumption fails

**VERDICT** 92W

Both groups counted layers as far as they could. Then they handed the rest to the same flow model, with the same assumption of uniform vertical strain. Below those counts the two age scales are not independent estimates that happen to agree; they are one calculation applied twice. What survives the model being wrong is the counted section, which is arithmetic on measured layers. The other is the tephra horizon, whose date came from a volcano and not from any ice. Those two are the comparison. The rest is a shared assumption.

**TAKEAWAY** 16W

Two numbers that agree because they came through the same step have not confirmed each other.

## 10.2 What the carbon dioxide change was worth

TRAPPED GAS · GAS LABORATORY · A ROOM · BALLPARK

**SCENE** 30W

Adeyemi has carbon dioxide at 190 parts per million through the cold interval and 280 at the top of the core. Halvorsen wants the difference expressed as an energy imbalance.

**GUIDE** 55W

Four numbers, and one of them is the isotope temperature slope, which belongs to another question. Ask of each whether this forcing uses it. And note the shape of the relationship. The bands are already partly saturated. Each extra molecule does less than the last, so the effect follows a logarithm rather than a proportion.

**ASK** 8W

Estimate the radiative forcing between the two concentrations.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

a greenhouse gas reduces the rate at which the surface loses heat to space

**VERDICT** 87W

Carbon dioxide absorbs in bands that are already partly saturated. So each extra molecule does less than the one before it. The forcing goes as the logarithm of the concentration rather than in proportion to it. That is why the effect is usually stated per doubling: every doubling adds about the same. Between the cold interval and the top of the core the concentration rose by half. That is worth about two watts on every square metre of the planet, sustained for as long as it lasts.

**TAKEAWAY** 12W

A concentration becomes a climate statement only through the energy it changes.

## 10.3 Half a per mil, in degrees

COLD LABORATORY · A PERSON · CHOICE

**SCENE** 52W

The 0.4 per mil laboratory-scale correction is in. The two isotope records still differ by about 0.5 per mil. That is with the cold interval paired on the published age labels. Tanabe wants that residual in the units of the temperature claim. Nobody should call it a different climate until it is.

**GUIDE** 66W

Four readings of half a per mil, and one of them refuses to convert at all. Ask of each whether it uses this site's own slope, and in which direction. Multiplying where you should divide gives an answer far too large. And note why the conversion matters. In per mil the difference sounds like rounding, and in degrees it is a quarter of the season's headline.

**ASK** 17W

What is half a per mil worth in degrees, before anybody claims it is a different climate?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

an isotope difference is converted to a temperature difference by the site slope · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

**VERDICT** 75W

Using Vestri's calibration, 0.5 per mil corresponds to about 0.75 degrees. That is large enough to matter to a roughly 2.7-degree cooling, but converting the size does not identify the cause. The isotope instrument measured the 0.5 per mil difference after the laboratory correction. The next question is whether the two curves are

> pairing the same moment in time. A chronology error can change which isotope samples are compared without changing either measured isotope value.
>
> **TAKEAWAY** 15W
>
> Convert the residual into the units of the claim, then keep magnitude separate from cause.

**END OF THE DAY** 15W

Two records do not independently confirm a result when the same assumption built both clocks.

DAY 11 OF 15

# The two clocks inside one core

**PLAN CARD** 38W

Thursday. Vestri's gas is about 450 years younger than its ice, and Skarv's is only 110. Today you remap the same air event onto each site's own ice ages, and see how much of the gap that removes.

**BRIEFING** 20W

The two sites have very different gas–ice age offsets, so the same atmospheric event was paired to different ice ages.

**WHAT YOU DECIDE** 15W

Test whether the remaining mismatch comes from pairing ice samples to the wrong atmospheric time.

## 11.1 Pair the same atmospheric event to the right ice

TRAPPED GAS · GAS LABORATORY · A ROOM · VERIFY

**SCENE** 62W

Adeyemi finds that the cold interval was paired between sites using a sharp atmospheric gas change. The nearby isotope samples were then given the gas age, as if gas and ice shared one clock. Vestri's estimated gas–ice offset is about 450 years; Skarv's is about 110. He can now remap the same gas event onto the correct ice ages at both sites.

**GUIDE** 63W

Write down how much isotope difference you expect to remain after each site gets its own gas–ice age correction. Then remap the same atmospheric event onto the ice chronology at Vestri and Skarv. Finally check the gas clocks against the sharp methane feature present in both cores. The check has to live in the gas record, because that is the clock being tested.

**ASK** 24W

After each site gets its own gas–ice age correction, how much isotope difference remains between the ice samples paired to the same methane change?

**BEHIND THE BUTTON** 137W

What changes and what does not. The isotope measurements themselves stay fixed. The correction changes the age assigned to the atmospheric gas event. That changes which ice-isotope samples are paired across the two sites. That distinction is the scientific heart of the day.

Why methane is the check. Abrupt methane changes are recorded in trapped air at many ice-core sites and can be used as gas-phase synchronization markers. A volcanic layer is an ice-phase marker. It is excellent for the ice clock, and the wrong independent check for a gas-clock correction.

Why the nominal answer is not the final claim. Δage depends on firn densification, accumulation, close-off, and the age distribution of trapped air. The best-fit correction can reveal a real chronology effect. It can still carry a range wide enough to matter to the final comparison.

**TAKES AS READ**

a correction should be predicted before it is applied, or it cannot be wrong

**VERDICT** 114W

Gas and ice at one depth do not share an age. Suppose a gas event known worldwide is used to line up two cores. Each site's own gas–ice age difference has to be applied first, before the matching ice-isotope samples are chosen. Vestri moves by much more than Skarv because its low accumulation leaves a larger offset. On the best estimate, that remapping reduces the 0.5 per mil residual to about 0.15. The independent check is a methane feature present in both gas records. Methane is well mixed, so an abrupt change in it can synchronise gas chronologies across cores. The remaining range is a chronology uncertainty, not a change to the isotope data.

**TAKEAWAY** 14W

A chronology correction changes which observations are compared; it never changes the observations themselves.

## 11.2 A clock that does not use the flow model

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · BALLPARK

**SCENE** 34W

Okonkwo has a radiocarbon measurement on particulate organic carbon extracted from one deep section. After blank correction, its activity is 0.55 of the modern reference. The conventional radiocarbon mean lifetime is on the bench.

**GUIDE** 98W

Five numbers, and three of them are traps. One is the half-life of the same isotope, which is not the same as its average life. One is the ice thickness. One is cosmogenic $^{10}$Be — beryllium made high in the air by cosmic rays, and the other clock down here. It is read by matching solar and magnetic events, not by decay, so it does not belong in this sum. Ask of each number whether this clock depends on it. The decay rate cares nothing about temperature, pressure or flow, so it owes nothing to the flow model.

**ASK** 26W

This section has two clocks the flow model did not build: decay, and the cosmogenic $^{10}$Be markers. Estimate its radiocarbon age, before calibration to calendar years.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

a radioactive isotope decays at a rate nothing in the ice can change

**VERDICT** 82W

Radiocarbon in carbon-bearing material trapped in ice decays independently of ice flow. Using the measured activity relative to a reference gives a conventional radiocarbon age of about 4,900 years. Real ice-core radiocarbon work also has to control tiny carbon blanks. And it has to turn radiocarbon years into calendar years, because atmospheric carbon-14 has varied. The value is useful here because none of those steps uses the deep flow model. On its own it is an outside constraint, not a replacement chronology.

**TAKEAWAY** 16W

An independent clock is valuable because it constrains the flow model, not because it is assumption-free.

## 11.3 What the nominal correction actually proves

RECORDS & REPORTING · SCIENCE MODULE · A PERSON · CHOICE

**SCENE** 36W

On the best-estimate clock correction, the paired isotope samples differ by only 0.15 per mil. But Adeyemi's Δage estimate still has a model range. The four firn sections meant to tighten it have not been re-measured.

**GUIDE** 64W

Four readings of what is left after the correction. Ask of each whether the remaining difference is bigger or smaller than the uncertainties either group quotes. And ask what a correction does to a result. This one was predicted before it was applied, then checked against a date that came from neither calculation. Both records were on the wrong clock; only one moved far.

**ASK** 6W

What does the result support tonight?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

a difference smaller than the uncertainty of either measurement is not a measured difference

**VERDICT** 68W

The best-fit remapping is strong evidence that chronology contributed to the apparent disagreement. It is not yet evidence that chronology explains all of it, because the gas–ice age difference is a model-constrained quantity with uncertainty. The scientifically defensible result tonight is therefore causal but limited: the clock mattered, and the size of what remains is still uncertain. Day 14 will ask which sentence survives the full credible range.

**TAKEAWAY** 16W

> A correction can identify a real cause before it is precise enough to close the case.

**END OF THE DAY** 17W

A clock correction changes which samples are comparable; it does not change what the isotope instrument measured.

# A day when nothing is wrong

**PLAN CARD** 54W

Friday. For the first time this season nothing is broken, and that makes the choice harder. Nine drilling days are left and four good jobs want them. Today you commit the last of the drill time. Some of that work would help the station, and only some of it can still change the record.

**BRIEFING** 19W

Good work and work this season needs are not the same list, and the fuel was committed before anybody

**WHAT YOU DECIDE** 13W

Spend the remaining season on measurements that can still change the final claim.

## 12.1 Nine days, four jobs

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · ALLOCATE

**SCENE** 31W

Lindqvist has nine drilling days before the station closes. Anneke de Vries, the camp manager, has the fuel committed for those nine and no more. Four pieces of work want them.

**GUIDE** 56W

Nine drilling days, fuel committed for exactly nine, and four jobs that want them. Watch the questions list rather than the depths: it shows which questions the current plan can still answer. Every day you spend is a question you can answer and one you have given up. Nothing can be added after the station closes.

**ASK** 9W

Which questions can the last nine days still answer?

**BEHIND THE BUTTON** 140W

Why the fuel is the real constraint. Days can in principle be found; the fuel for those days cannot. It was committed before the season and there is no more. The plan has to fit the fuel rather than the calendar.

Why depth is not the same as value. Drilling deeper always produces more core, and the extra core answers a question only if it reaches something the record needs. A shallower job that closes a comparison can be worth more. A deeper one that extends a series nobody is arguing about is worth less.

What the closing date does to the choice. Anything unfinished is unfinished for a year. A core left mid-hole can be lost to closure, or to the hole itself. So a plan that just fits is worse than one that fits with something in reserve.

**TAKES AS READ**

a season ends on a date fixed by aircraft rather than by science · why ice keeps a record at all: burial without melting — taken as read

**VERDICT** 96W

The two required questions are the ones this season exists to answer. Whether the deep section reaches older ice. And whether the horizon everything is now tied to appears in ice that shared the weather but not the drill. Together with the fluid change they use exactly the nine days available. The firn profile and the borehole log are both good work. Both can be done next season by somebody who has not yet drilled to 2,400 metres. A plan that fits everything in is a plan that has not been asked what it is for.

**TAKEAWAY** 18W

A plan is judged by which questions it can still answer, not by how much it fits in.

## 12.2 The stake that gained a centimetre

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · CHOICE

**SCENE** 29W

Brandt has 48 stakes read this morning. One near the hut has gained a centimetre since the last reading. 11 others across the array have lost about the same.

**GUIDE** 65W

Four readings of one stake that gained a centimetre while eleven others lost. Ask of each whether a single stake can separate snow that fell from snow that arrived sideways. Wind moves far more snow across this plateau in a fortnight than falls out of the sky. Summed across the array, the moving snow cancels. That is why forty-eight stakes are read rather than one.

**ASK** 9W

What does the gain at that one stake establish?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

wind moves snow across a surface after it has fallen

**VERDICT** 80W

Wind on this plateau moves far more snow horizontally than falls out of the sky in a fortnight. So any single stake is measuring snowfall and redistribution added together. It cannot separate them. The array can. Snow moved from one place to another cancels when the whole array is summed. What is left is what actually arrived. That is why 48 stakes are read rather than 1, and why the accumulation figure is a season's arithmetic rather than a reading.

**TAKEAWAY** 18W

A measurement of a surface that moves has to be made in enough places to catch what moved.

## 12.3 The order a run is made in

COLD LABORATORY · A PERSON · SEQUENCE

**SCENE** 29W

Cruz has 16 samples to run and the instrument warmed up. Tanabe's rule is that a run has to be able to say afterwards whether it can be trusted.

**GUIDE** 61W

All five of these will be done, so ask what each one makes possible. A blank says what the process adds with no ice in it. An opening standard fixes the scale. Standards inside the run reveal drift while it happens. Ask which of them a correction could be computed without. And note why both raw and corrected values are kept.

**ASK** 6W

Fix the order of a run

**BEHIND THE BUTTON** 169W

Why the order is graded whole. A sequence is a claim about dependency. Each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits. Partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them. That usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply. What is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility. Reversibility puts every observation that leaves the sample as it was before any that consumes it. On those boards a chronological reading finds nothing. None of the steps has to happen at a particular hour.

**TAKES AS READ**

an instrument's response can move slowly during a long run

**VERDICT** 78W

Every step here exists to make the run auditable. The blank establishes what the process adds before any ice is involved. The opening standard fixes the scale. The standards inside the run reveal drift while it is happening. The closing standard measures how far it moved overall. Only then can the samples be corrected. Both the raw and the corrected values are kept. A correction is a claim about the instrument that somebody may want to check later.

**TAKEAWAY** 17W

A run is designed so that it can be checked afterwards, not so that it finishes soonest.

**END OF THE DAY** 13W

When time is scarce, fund the measurement that can still change the conclusion.

DAY 13 OF 15

# The night the generator stopped

BEFORE THE DAY — CANVASS

### Who actually heard the generator stop

The number two generator stopped some time in the night and the log says nothing about it. How long the core store was without power decides what the season can claim. Ask round the camp until enough accounts agree on when the lights flickered.

**PLAN CARD** 52W

Saturday. The number-two generator died in the night, and the store came back at minus 19 instead of minus 20. Four gas sections can no longer be called clean. Today you say what the broken record actually spoils. Then you restart what is still drifting, and put a number on the scatter.

**BRIEFING** 16W

The core store crossed its validated handling limit overnight and the log does not say when.

**WHAT YOU DECIDE** 17W

Decide what the cold-chain excursion actually invalidates, what must be quarantined, and what uncertainty it leaves behind.

## 13.1 Six signatures and two hours

COLD LABORATORY · A ROOM · ATTEST

**SCENE** 24W

Six lines on the night sheet, all signed. Tanabe has two hours before the flight manifest closes and can verify two of them properly.

**GUIDE** 56W

Six signed lines, two hours, and two of them can be verified properly. Open each claim and read what already backs it. A signature records that somebody took responsibility, not that anybody observed the thing. Hold what the backing does not support. Spend the two hours where being wrong would matter after the flight has gone.

**ASK** 8W

Which two claims should the two hours verify?

**BEHIND THE BUTTON** 139W

Why the manifest deadline is the whole pressure. Once the flight closes, anything unverified stays unverified until next season. The samples or the equipment it concerns are already off the ice. A claim checked next year is a claim about something nobody can revisit.

What a night sheet is. A running record of what was done on shift, signed at the end. It is written by people who were tired, at the end of a working night. Its purpose is continuity rather than evidence. That is exactly why some lines on it are firmer than others.

How to choose between two doubtful lines. Not by which is least likely to be true. By which one, if wrong, would change what somebody does with the material afterwards. That is consequence, and it is the judgement the two hours are for.

**TAKES AS READ**

a signature records that somebody believed something, at the time they signed · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

**VERDICT** 79W

The signatures prove that somebody recorded the claims, not that the conditions were continuously observed. The store was outside the laboratory's validated handling range when power returned, and the logger cannot say when the excursion began. That does not prove the gas measurements are ruined. It means the four chronology sections must be quarantined or specially tested rather than silently treated as ordinary samples. Because those sections were intended to tighten Δage, the break widens the final chronology uncertainty.

**TAKEAWAY** 20W

A protocol excursion creates a documented uncertainty; it does not automatically create either a valid sample or a destroyed one.

## 13.2 What comes back first

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · TRIAGE

**SCENE** 29W

Power is back on one generator. Lindqvist, de Vries and Tanabe each want their own system restarted first. The single generator will not carry all of them at once.

**GUIDE** 60W

Four systems want the one generator. Ask which condition is still moving away from a validated state while you wait. The core store is already outside its qualified handling range and is still warming. The drill loses schedule time, the chemistry queue can wait, and the satellite has another window. Restart what is still changing, not what is merely delayed.

**ASK** 4W

What is restarted first?

**BEHIND THE BUTTON** 129W

Why first is different from most important. Restart order depends on what another hour changes. A major system that will be in the same condition an hour from now can wait. A smaller system whose evidence quality is still drifting cannot.

Why the temperature boundary is procedural rather than magical. The laboratory validated its handling and gas-analysis method for sections kept at or below a stated temperature. Crossing that boundary creates a qualification problem, not an instant physical verdict. The correct response is to restore control and document the excursion.

Why the science still has stakes. These four sections were intended to tighten the gas–ice age estimate. Losing confidence in them does not erase the season's result. It widens the uncertainty of the correction the final comparison depends on.

**TAKES AS READ**

a system that has already failed is more informative than one that has not yet been used

**VERDICT** 94W

The store is still warming, so delay continues to change the condition of irreplaceable samples. The drill loses schedule time if it waits. The chemistry queue is still measurable tomorrow, and the satellite link can wait for the next window. Restart priority follows the rate at which evidence is becoming harder to defend, not the loudest request. The point is not that minus 20 is a universal physical cliff. It is the station's validated handling boundary. The longer the excursion runs, the less well those sections match the conditions the method was qualified under.

**TAKEAWAY** 12W

Restart what is still changing irreversibly or leaving the validated evidence envelope.

## 13.3 How much of it is noise

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · BALLPARK

**SCENE** 29W

Aalto has 30 years of reconstructed annual temperature from the top of the core. The year-to-year scatter is 1.4 degrees and the summary wants a mean for the period.

**GUIDE** 56W

Four numbers, and two of them belong to other questions: the calibration slope, and the years of mast overlap. Ask of each whether this uncertainty depends on it. Averaging does not remove the scatter of individual years. It reduces the uncertainty of the mean, and only as the square root of how many years went in.

**ASK** 12W

Estimate the standard error of the thirty-year mean, given the year-to-year noise.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

averaging independent years reduces the scatter of the average, not of the years · random against systematic uncertainty — taken as read

**VERDICT** 64W

Treat the year-to-year scatter as independent random variation. Then 1.4 degrees over 30 years gives a standard error of about 0.26 degrees on the mean. That calculation describes only the random component. A common isotope-temperature calibration error or a chronology shift moves many years together and does not average down with more years. The final uncertainty therefore cannot be built by quoting 0.26 alone.

**TAKEAWAY** 17W

Averaging shrinks independent random scatter; it does not shrink a systematic that moves the whole record together.

**END OF THE DAY** 17W

When the evidence chain breaks, record the break; do not convert uncertainty into certainty in either direction.

# What survives the uncertainty

**PLAN CARD** 34W

Sunday. Matching each site's two clocks leaves about 0.15 per mil between the records, and the honest range runs out to 0.4. Today you pick the strongest sentence that holds across all of it.

**BRIEFING** 19W

The nominal alignment removed most of the mismatch, but the generator incident left the gas-age correction wider than planned.

**WHAT YOU DECIDE** 15W

Choose the comparison statement that remains true across the credible range of the clock correction.

## 14.1 Four sentences and a range

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · STRESS

**SCENE** 48W

After the laboratory correction, 0.5 per mil remained between the isotope records. The best clock remapping removes about 0.35. The credible correction runs from 0.10 to 0.40 per mil. The firn model is uncertain, and four chronology sections are quarantined. Four possible report sentences are on the board.

**GUIDE** 61W

The correction removes somewhere between a tenth and four tenths of a per mil. Nobody can settle which, this season. Move it to the least favourable end before choosing between the four sentences. A sentence can be true at the middle of the range and false at the end of it. That is not a sentence the season summary can carry.

**ASK** 15W

Which sentence is still defensible across the full credible range of the systematic clock correction?

**BEHIND THE BUTTON** 138W

Why the pessimistic end rather than the best estimate. A published sentence is read as a claim about the world, not about the middle of somebody's uncertainty. Choosing at the expected value means publishing something that has a real chance of being wrong. The retraction costs more than the caution would have.

What makes one sentence survive and another not. Sentences that assert a direction usually survive. Sentences that assert a size usually do not. The size is exactly what the correction's range moves. A carefully weakened claim can be both true and useful.

Why this cannot wait for the assumptions to settle. The summary is written this season and the assumptions will not be settled for years. The professional move is to write the strongest sentence that survives the whole range. Not the strongest sentence full stop.

**TAKES AS READ**

a correction estimated from a model carries a range rather than a single value

**VERDICT** 89W

At the best estimate the chronology correction leaves only about 0.15 per mil. The final sentence has to survive the whole credible range, not the preferred point. If the correction removes only 0.10, a 0.40 per mil residual remains; if it removes 0.40, only 0.10 remains. That range is too broad to claim the records are fully consistent. It is also too broad to claim the sites cooled differently. The robust result is that chronology accounts for part of the original mismatch and the remainder is unresolved this season.

**TAKEAWAY** 20W

The claim to publish is the one that is still true at the least favourable end of what you assumed.

## 14.2 What the bubbles license

TRAPPED GAS · GAS LABORATORY · A PERSON · CHOICE

**SCENE** 31W

Adeyemi has carbon dioxide through the cold interval and the modern section, measured on this core and reproduced on the replicate. Halvorsen wants to know what it supports on its own.

**GUIDE** 65W

Four options, and the question is what this measurement is of. Ask of each how far the quantity travels before it is measured. Carbon dioxide stays in the air long enough to mix through all of it, so a bubble here reports the planet. The isotopes record condensation over this region, and reach degrees through a slope fitted at this site. Local at both steps.

**ASK** 12W

What does the carbon dioxide record support that the isotope record cannot?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

a measurement made at one site can record a quantity that is the same everywhere

**VERDICT** 79W

Carbon dioxide has a long atmospheric lifetime and mixes fast. So ice-core concentrations stand for the whole atmosphere rather than for the weather over Vestri. The measurement therefore supports a statement about atmospheric composition at the gas age assigned to the sample. Water isotopes answer a different question: they are a local-to-regional precipitation proxy whose conversion to temperature needs a site calibration. Strong gas concentration data can still have a less certain date if the gas chronology is uncertain.

**TAKEAWAY** 19W

> Some quantities measured at one place are global, and a proxy of local weather is not one of them.

## 14.3 Before anything is sent

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · SEQUENCE

**SCENE** 25W

Okonkwo has the sentence, the corrected records and a report that closes in the morning. Four things are outstanding and their order is not obvious.

**GUIDE** 66W

All five of these will be done before the report goes. Ask of each what the next step would mean without it. Nobody can reproduce a result without knowing which version of the age scales produced it. Recomputing from raw data tests that the draft's number is the method's number. And one of these turns a disagreement about another group's record into a conversation with them.

**ASK** 7W

Fix what happens before the report goes

**BEHIND THE BUTTON** 169W

Why the order is graded whole. A sequence is a claim about dependency. Each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits. Partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them. That usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply. What is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility. Reversibility puts every observation that leaves the sample as it was before any that consumes it. On those boards a chronological reading finds nothing. None of the steps has to happen at a particular hour.

**TAKES AS READ**

a result should be reproducible by somebody who was not present

**VERDICT** 91W

Each step is what makes the next one mean something. Freezing the age scales fixes what the result was computed on, which is the only way anybody can reproduce it later. Recomputing from raw data on that frozen version tests that the number in the draft is the number the method produces. Writing the correction and its range into the methods is what lets a reader disagree with it specifically. Circulating to the other group is what turns a disagreement about their record into a conversation with them. Then it goes.

**TAKEAWAY** 16W

A result leaves the station as a package: the numbers, the version, and what was assumed.

END OF THE DAY 13W

A publishable claim must survive the uncertainty in the assumptions that produced it.

# Sign the record

**PLAN CARD** 34W

Monday. The report is written and the aircraft leaves on Wednesday. Today you sort what was measured from what was guessed, then sign — or refuse to sign — the sentence others will quote.

**BRIEFING** 17W

The Vestri Record is complete; the last job is deciding exactly what you are willing to sign.

**WHAT YOU DECIDE** 11W

Separate measurement, calibration, model, and inference—then make the final release decision.

## 15.1 What rests on what

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · PROTOCOL

**SCENE** 44W

Brandt has the season's headline quantities on the wall. Before you sign anything, each one has to be labelled by how it was obtained. The four labels are direct measurement, calibrated proxy, counted or modelled chronology, and atmospheric measurement on its own age model.

**GUIDE** 63W

Four headline numbers, and four descriptions of how each was got. Pair them by asking how many steps sit between the instrument and the number. One is a direct surface measurement checkable next year. One passes through a slope fitted to eleven years. One is counted where counting worked and modelled below. And one is a solid measurement attached to a modelled date.

**ASK** 6W

Sort measurement, calibration, model, and inference

**BEHIND THE BUTTON** 193W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison. The question is not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. The responses are a set to be distributed rather than a list to be sampled. So every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation. That is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

**TAKES AS READ**

a number is only as strong as the weakest step between the ice and it

**VERDICT** 86W

The accumulation is observed at the surface and converted to ice equivalent. The cooling comes from measured water isotopes passed through a site calibration. The chronology is counted where annual layers are resolvable and modelled below that, with volcanic and radiocarbon constraints. Carbon dioxide is measured directly in trapped air. The age given to that air belongs to the gas chronology, not to the ice depth alone. Those categories are part of the result because they tell the next reader which uncertainty can move which conclusion.

**TAKEAWAY** 18W

Every number in a proxy record belongs to a class, and the class is part of the number.

### 15.2 The shortest thing this ice can show

COLD LABORATORY · A PERSON · BALLPARK

**SCENE** 39W

Tanabe has the firn diffusion length and the annual layer thickness at deposition. Before the final sign-off, the next team wants one number. It is the shortest surface event the isotope archive could keep, before deep-flow thinning is considered.

**GUIDE** 75W

Use the diffusion length and the surface annual layer thickness only. Twice the diffusion length is a rule-of-thumb width below which a feature is strongly smoothed in firn. Dividing that width by one year's deposited thickness gives a near-surface timescale. Do not use the deep layer thickness here. Once the ice has flowed and thinned, the diffusion profile has been thinned with it. The deep resolution then needs a forward model rather than this shortcut.

**ASK** 13W

Estimate the near-surface event duration that corresponds to twice the firn diffusion length.

**BEHIND THE BUTTON** 103W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated. Nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs. Then, whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit mismatch before it is placed. It also catches a quantity belonging to a different part of the problem. That is the habit this format exists to build.

**TAKES AS READ**

water molecules move through firn before the pores seal, averaging nearby layers together

**VERDICT** 84W

Water-isotope variations are smoothed by molecular diffusion while the snow is still firn. Using twice the diffusion length as a simple width criterion gives 0.16 metres. Dividing by the 0.119-metre annual layer at deposition gives about 1.3 years. This is deliberately a near-surface estimate. Deeper in the core, ice flow changes both the annual layers and the diffusion profile they inherited. So the time resolution has to be carried through a thinning-and-diffusion model. Reusing the surface diffusion length unchanged does not work down there.

**TAKEAWAY** 21W

A simple resolution estimate is valid only on the depth scale for which its diffusion length and layer thickness are defined.

## 15.3 Sign the Vestri Record

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

**SCENE** 54W

The final page is on Halvorsen's screen. The Vestri cooling estimate is ready and the laboratory-scale offset with Skarv is measured. The chronology correction is real but uncertain, and the deep ages are still model-dependent. Four dispositions are possible. Only one says everything the evidence supports without claiming what the season did not settle.

**GUIDE** 57W

Four sentences, all of them true in some sense. Ask of each what it claims beyond what was measured here. One calls two records independent, and below the counts they share a flow model. One rounds until the uncertainty disappears. One is a claim about every record at this latitude, and nobody here has read them all.

**ASK** 4W

What do you sign?

**BEHIND THE BUTTON** 197W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is. The right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Before it was fixed, that was 88 per cent of passage quizzes here. It is now checked per game as a binomial tail. Length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them. Each is the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That holds until the day the approximation is what is being tested.

**TAKES AS READ**

a claim is supported when the evidence would have come out differently if it were false

**VERDICT** 107W

The record has enough evidence to be useful and not enough evidence to support the strongest possible story. Vestri's accumulation, isotope measurements, gas concentrations, counted chronology, and independent tie points all survive the audit. The deep ages remain model-dependent. The 0.4 per mil laboratory offset is measured. The gas–ice clock mismatch is a demonstrated contributor to the gap that is left. The size of the final residual is still not known tightly enough. It cannot be

> called full agreement, and it cannot be called different climates. Signing the qualified release preserves the result and the uncertainty together. That is the scientific decision the fifteen days were for.
>
> **TAKEAWAY** 15W
>
> The strongest defensible conclusion is the one that keeps the measurement and its limits together.

**END OF THE DAY** 18W

The final scientific act is not another calculation. It is deciding what the evidence licenses you to say.

## THE CLOSING CARD

At 08:17 on Monday the Vestri Record went out. The first mismatch with Skarv had been about nine tenths of a per mil. Four tenths belonged to the two laboratories' reporting scales. Much of what remained came from pairing an atmospheric gas event to ice on two different gas–ice clocks. The best alignment left only a small residual, but the final correction was not known tightly enough to turn that best value into a claim that the two sites were identical. The signed sentence said what survived the whole range: chronology explained part of the apparent disagreement, and the remainder was unresolved.

The hole closed the season near 2,470 metres. Annual layers could be counted only through the section where the signals stayed resolvable; below that, the age scale remained a flow model constrained by volcanic and radiocarbon evidence. The generator failure did not magically destroy four gas sections. It broke their validated cold-chain record, so they were quarantined and the gas–ice correction stayed wider than planned. That uncertainty is printed in the record instead of disappearing into it.

The aircraft lifted with the Skarv material on board and the two groups still had a scientific question between them. They did not have a false answer. You measured the common scale while you still could, kept the gas and ice clocks separate, checked modelled ages against markers the model did not create, and signed the qualified sentence when a cleaner headline was available. The Vestri Record can now be used by somebody who knows exactly where it is strong and exactly where it can still move.

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Card text only — no options, boards or answers. 11,256 words of card prose across 52 stops. Generated from `themes/icecore` by `tools/make-cardbook.mjs`.